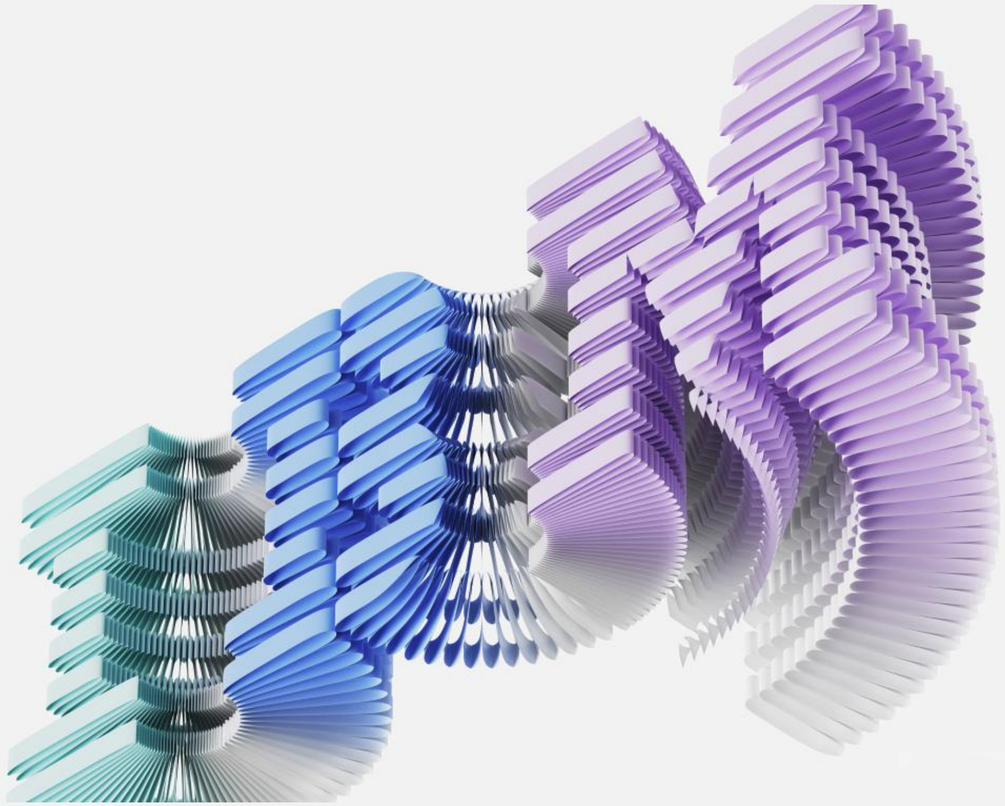
Realtime Data for Realtime AI



Nguyen Tuan Khang, *PhD*
IBM ASEAN

Moment of Intent

5min 8s 1s

Moment of Intent

5min
Target for digital loan approval

8s
Fraud prevention

1s
Response time

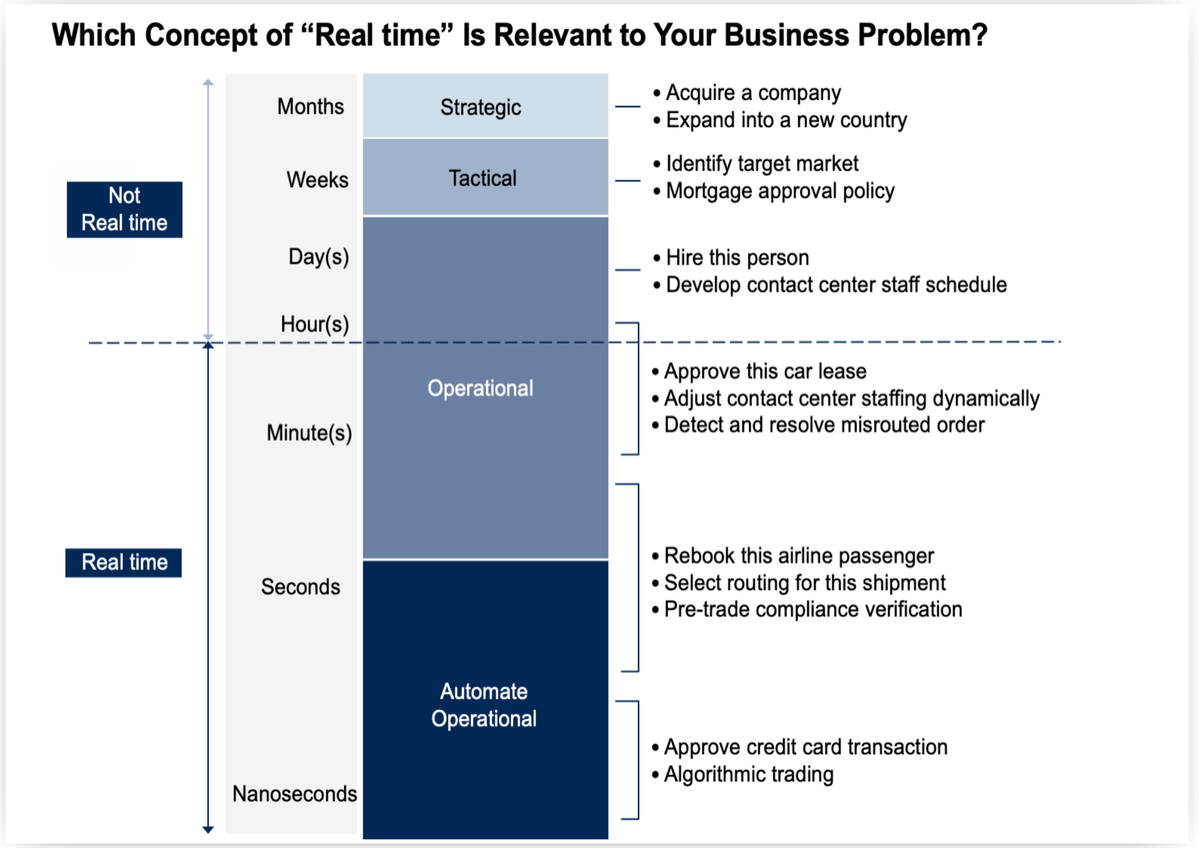
Concept of Real-time

Realtime is not only about response time



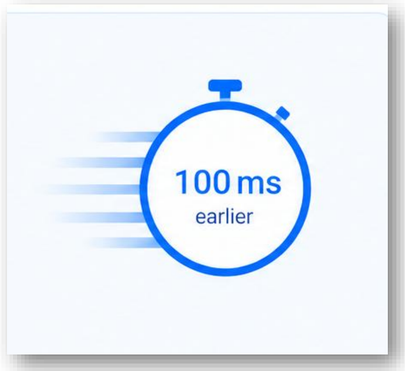
Something important happens in business and what we should do next?

It is about the Business Moment

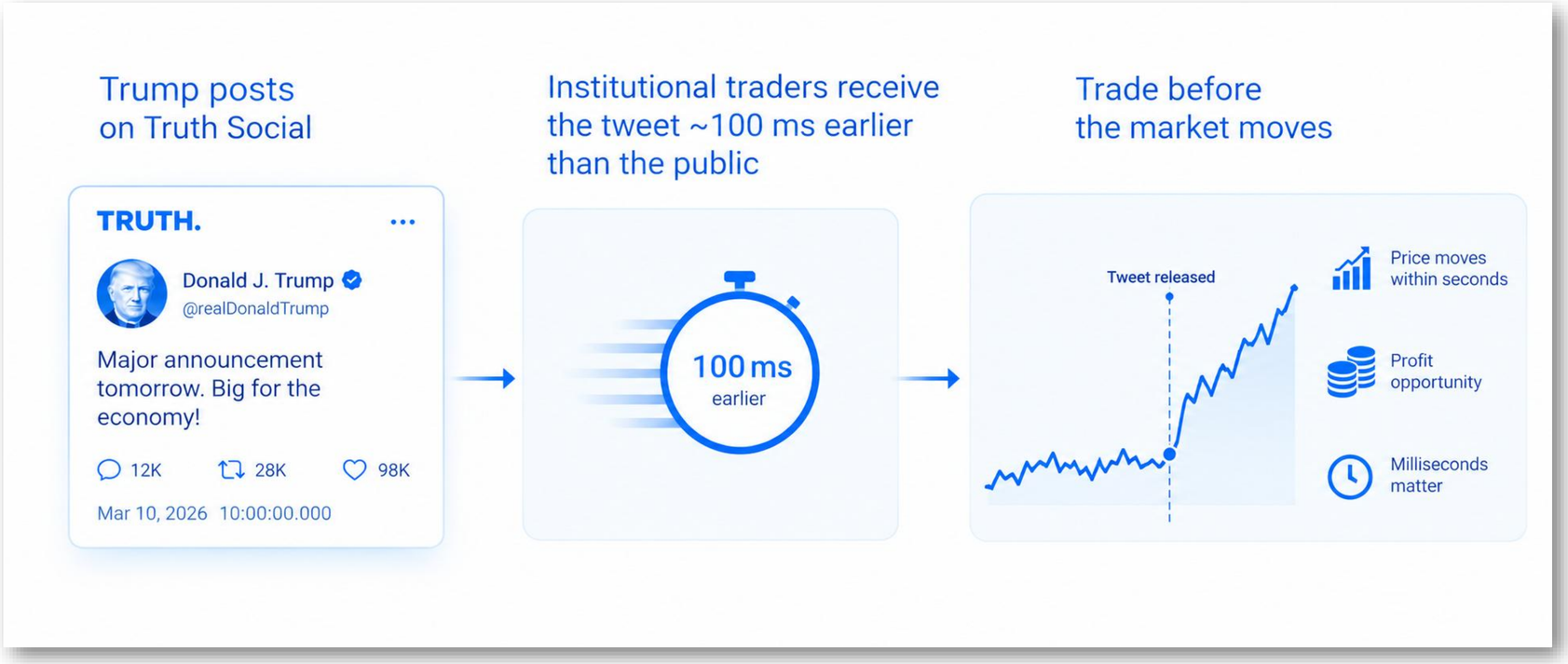


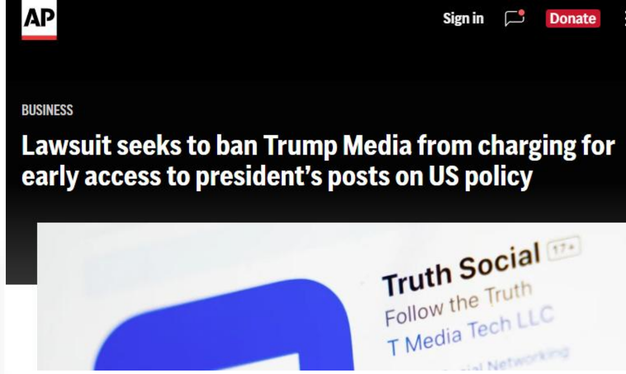
Source: Gartner, October 2025

Real-time Story



Real-time Story





Gartner

>60%

of AI projects that aren't supported by an AI-ready data practices will fail thru 2026

Gartner

26%

of CDOs are confident their data capabilities can support new AI-enabled revenue stream

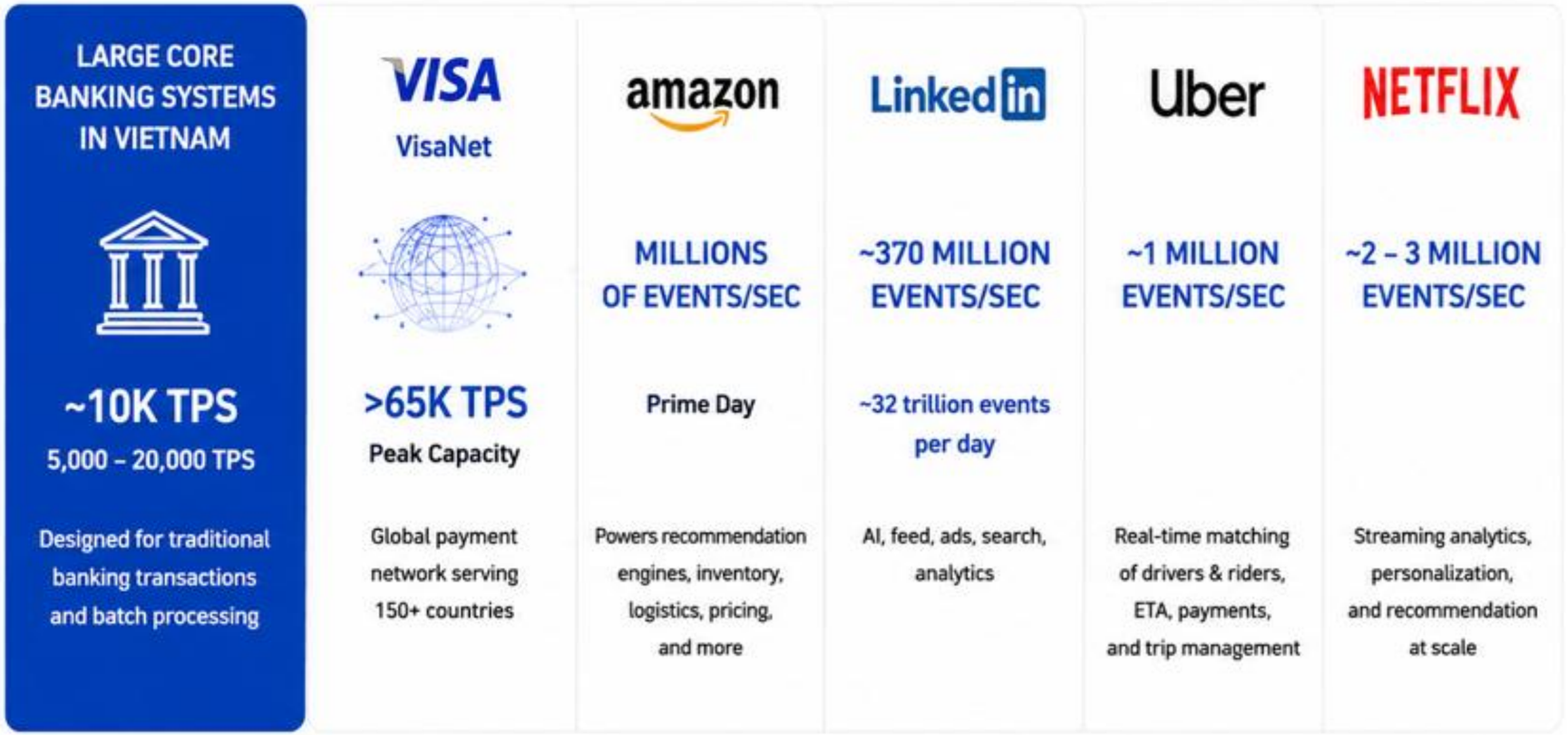
tdwi

77%

of companies are making analytics decisions on non-real-time data

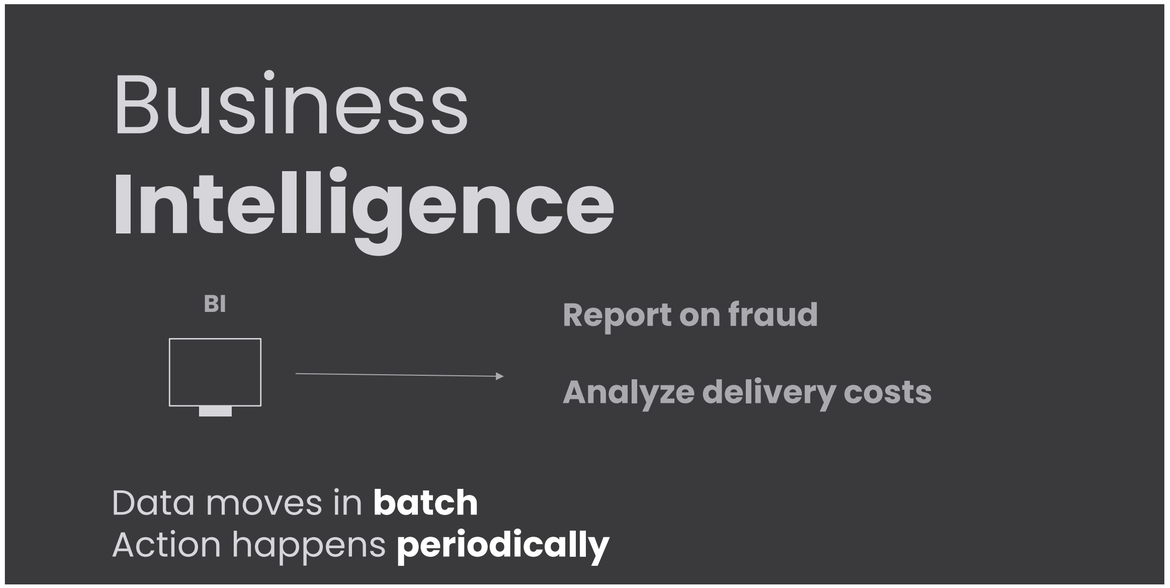
Reducing Time to Insight and Maximizing the Benefits of Real-Time Data, TDWI, 2024

Real-time Data for Real-time Business

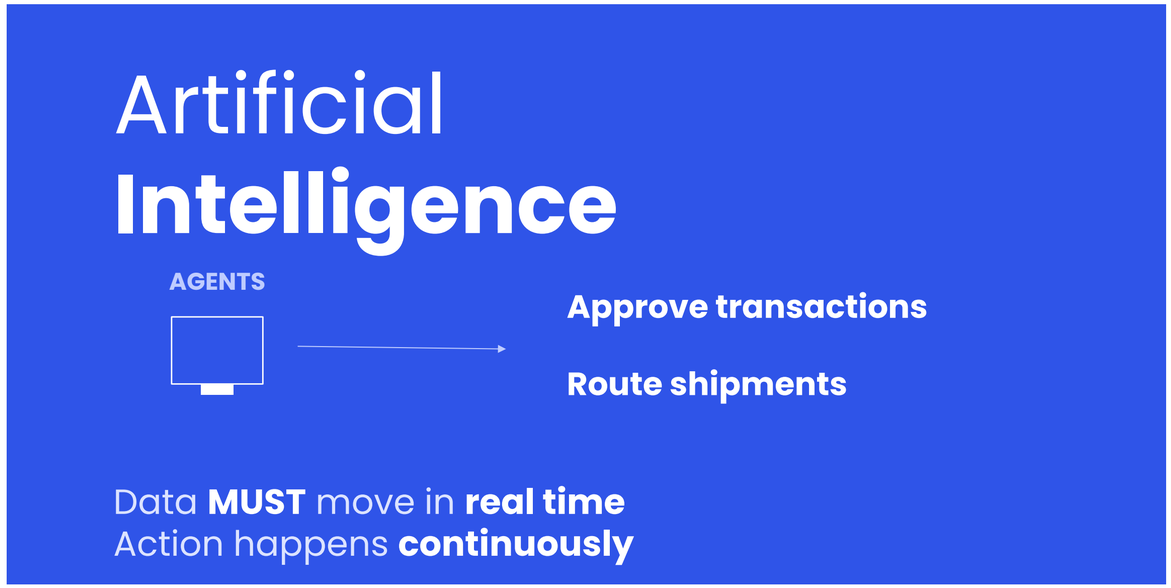


Data Streaming to move Data at real-time

Yesterday, software used by Human



Now, software used by Agents



- Build architectures for real-time data and data products.
- Data streaming is essential for real-time processing across decision intelligence, Agentic AI.

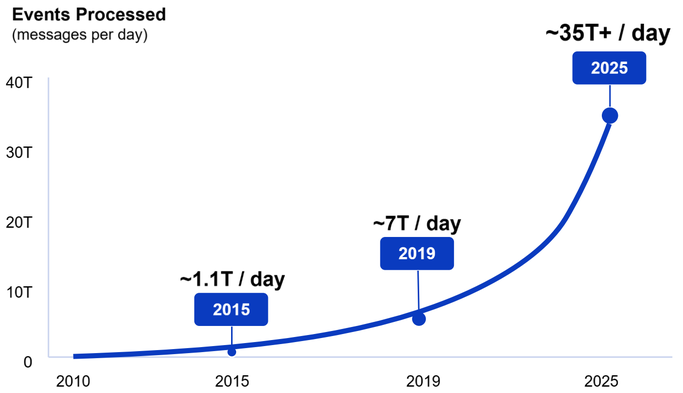
Source: Gartner, October 2025

LinkedIn Usecase with Event Streaming & AI

LinkedIn invented Kafka in 2010 to capture, distribute, and process billions of events in real time across the enterprise.

Workload Exploded

From billions to tens of trillions of events every day



~17 PB / day
of data ingested

Millions of Events / Second
at peak

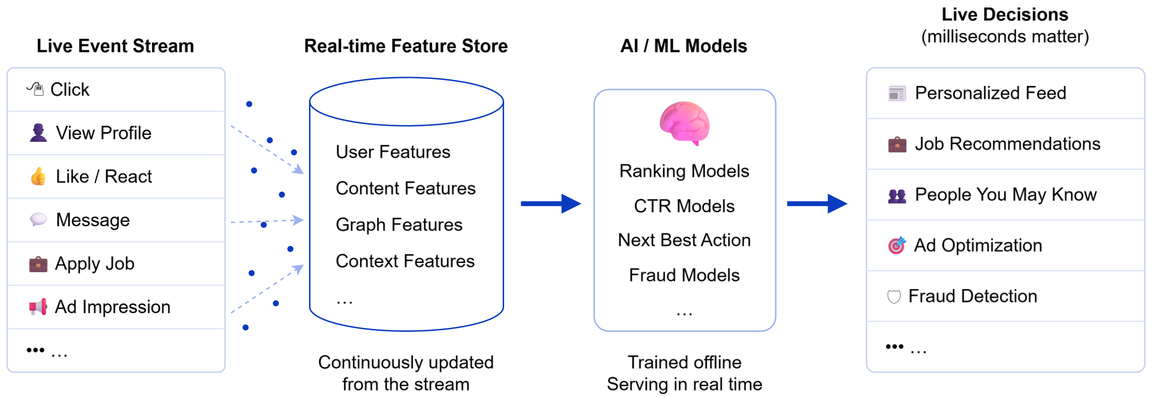
Global scale
across regions

Always on
24x7

Scale wasn't planned. It was earned—one event at a time.

Real-time AI, Real Impact

Events → Features → Models → Decisions in milliseconds

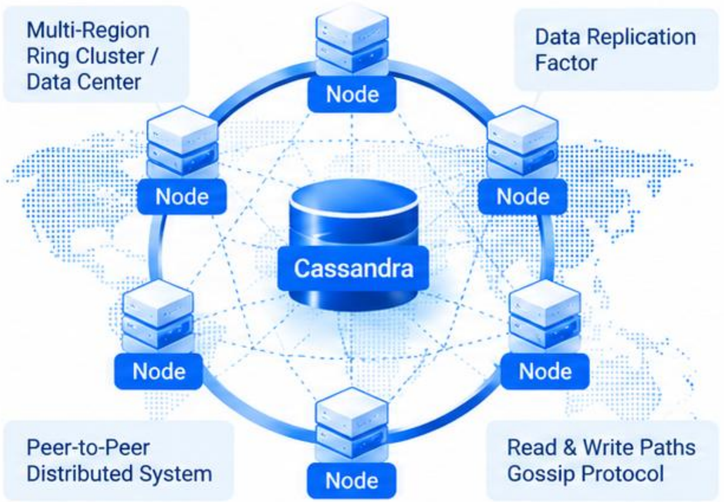


Every interaction is an opportunity. Real-time AI makes it count.

Platform for Realtime Data

Distributed Database

Scale-out, always-on, global

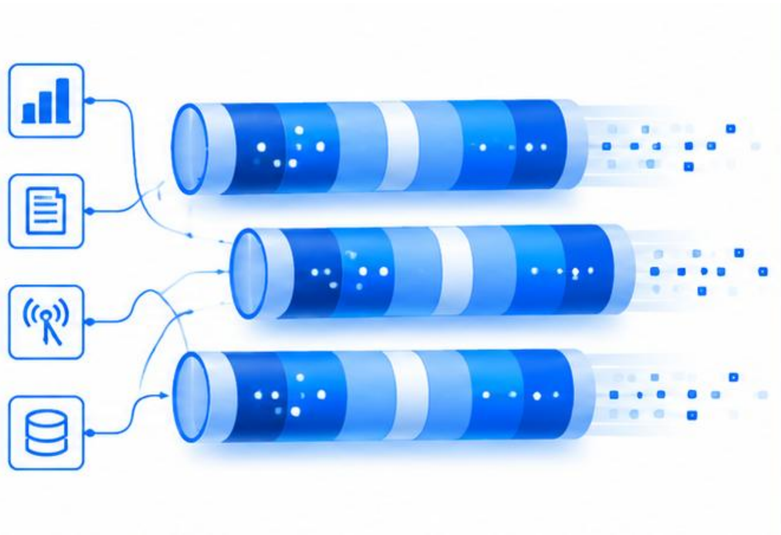


Benefits

- Horizontally scalable architecture
- High availability and fault tolerance
- Peer-to-peer distributed system

Event Streaming

High-throughput, real-time data pipeline

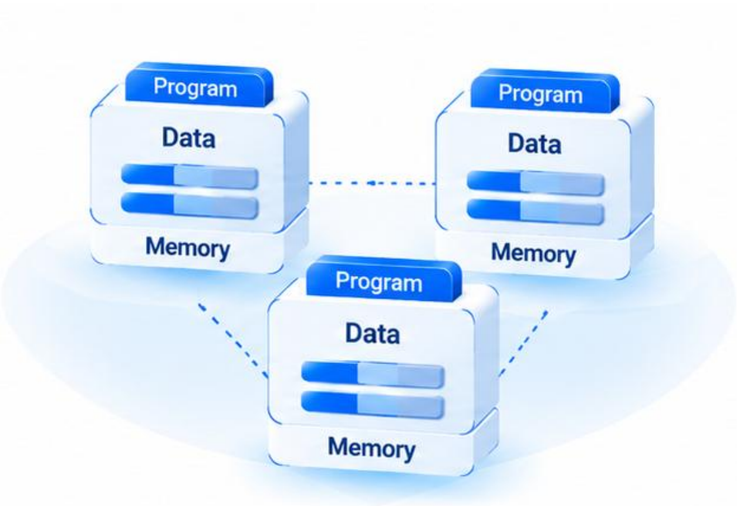


Benefits

- Real-time, continuous data ingestion
- High-throughput and low latency
- Scalable and reliable event pipeline

Memory-Optimized Processing

Fast, real-time, in-memory compute



Benefits

- Data processed in memory
- Low latency, real-time analytics
- High performance for modern applications

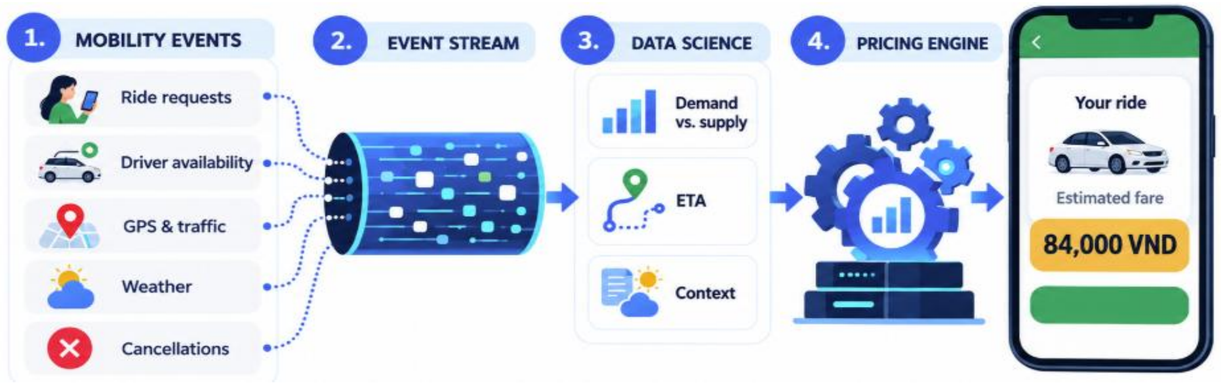
Case Studies: from Live Events to Realtime Business



Business Moment: Fresh price + real-time availability

- Billions of searches/days
- 1.5M price updates/min

https://medium.com/agoda-engineering/how-agoda-manages-1-5-trillion-events-per-day-on-kafka-f0a27fc32ecb

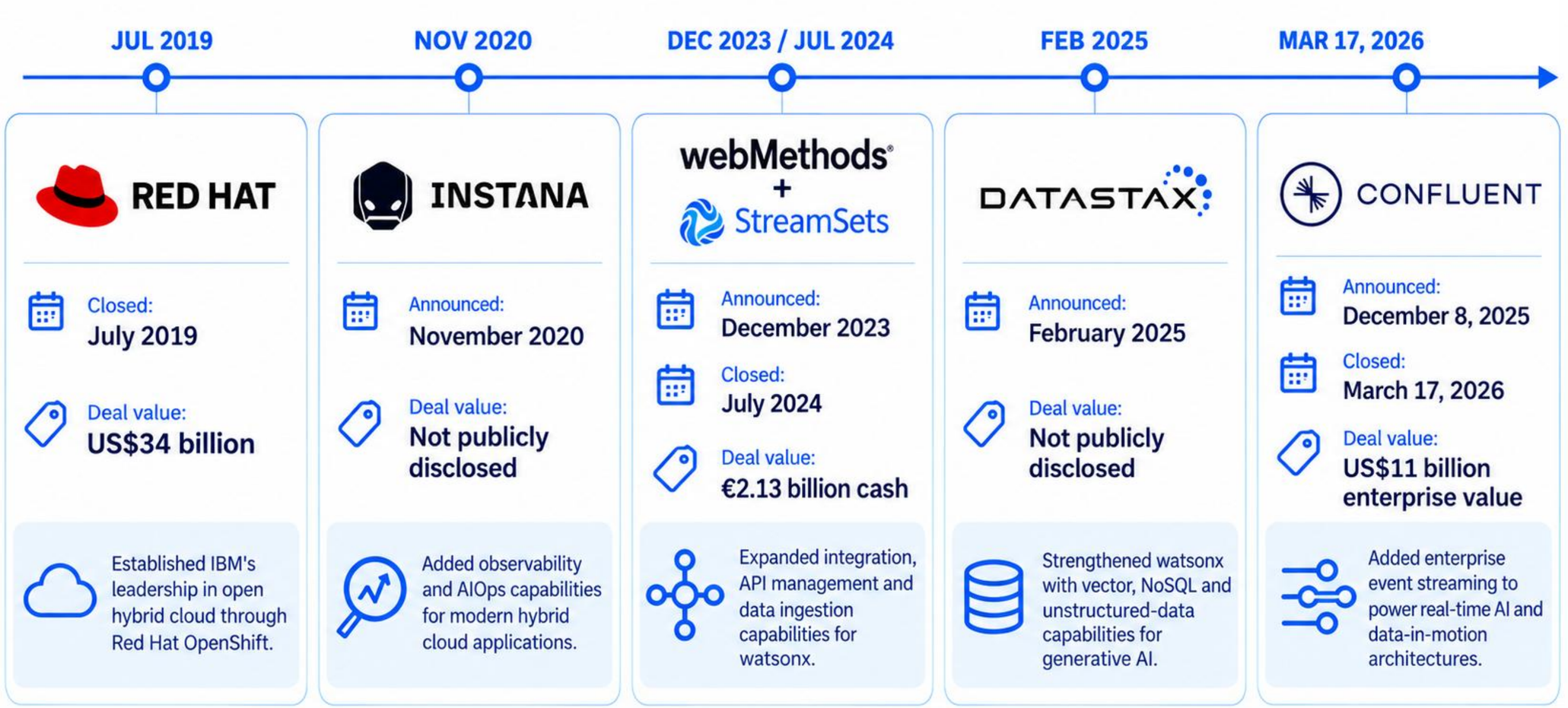


Business Moment: Real-time pricing

- 400k EPS
- 3.5M rides/day

https://engineering.grab.com/experimentation-platform-data-pipeline

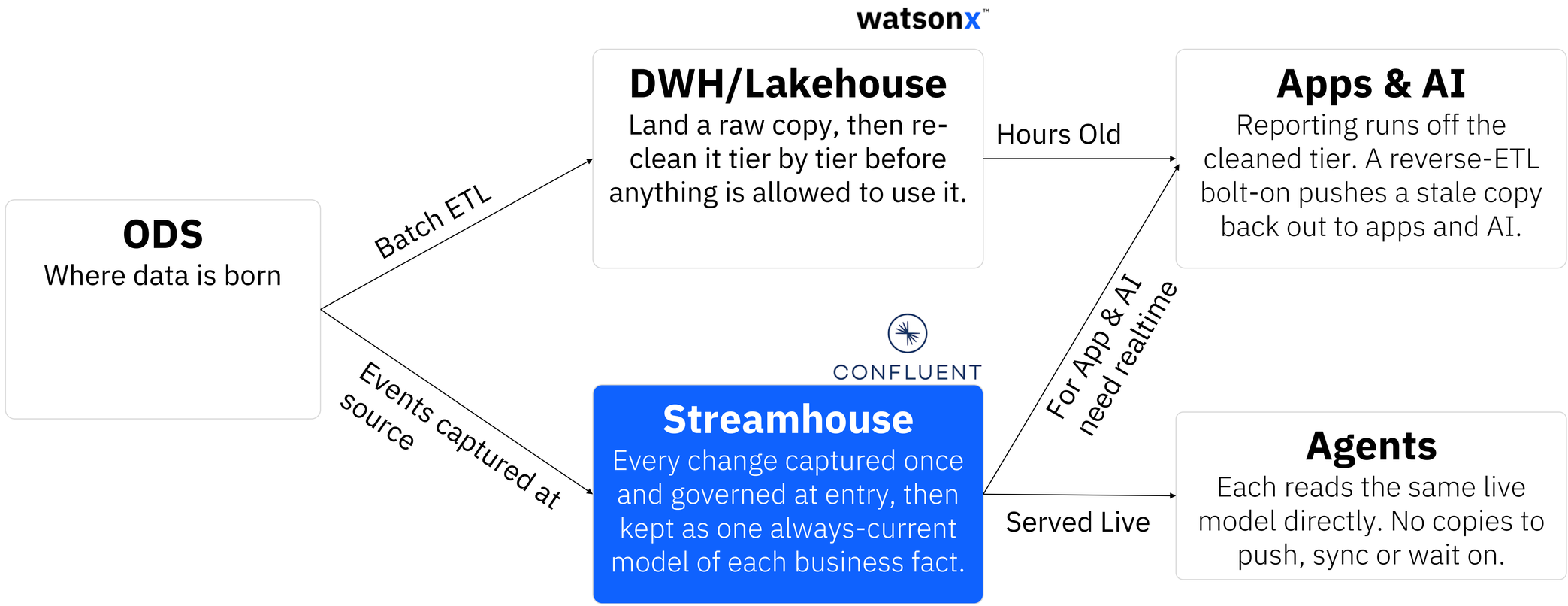
How IBM bet on Real-time Data?



Traditional Lakehouse



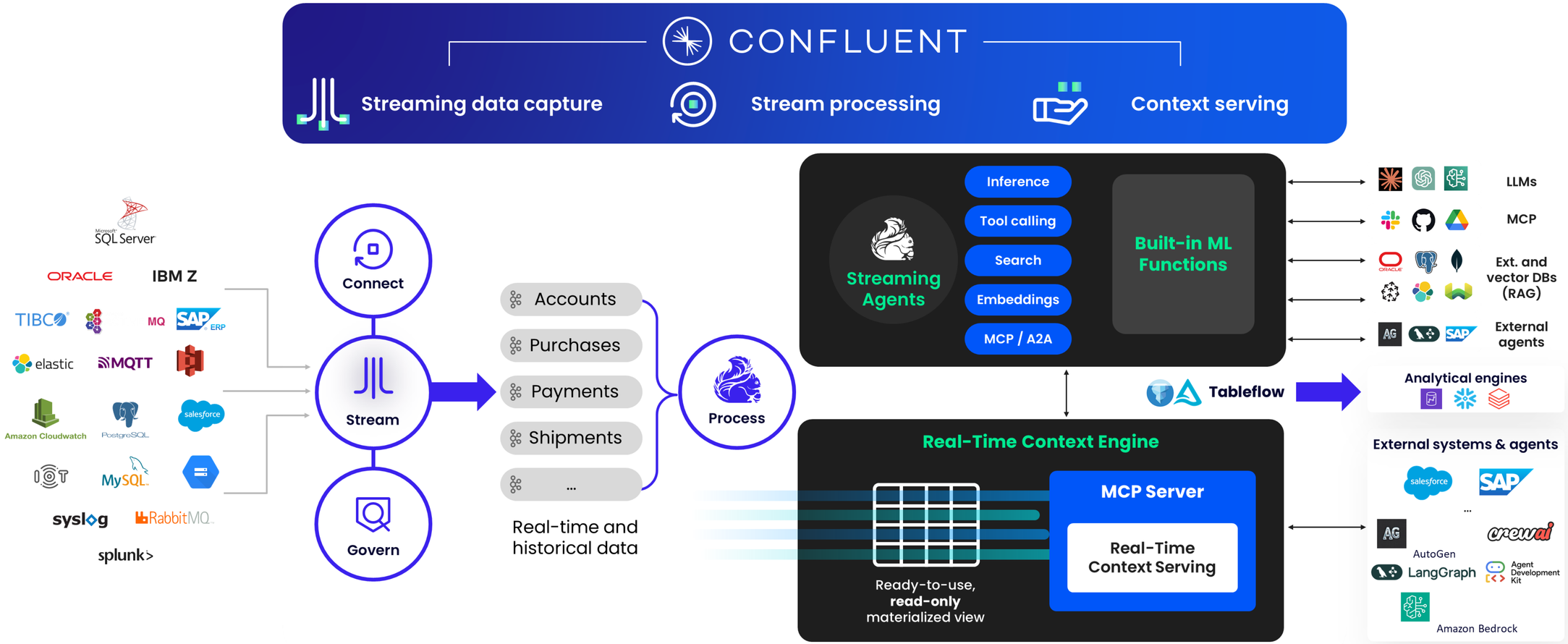
Streamhouse



A Streamhouse is a data architecture, not a product.

It continuously captures, processes and serves the current state of the business to applications, agents and analytics, so they all run on the same fresh, governed source of truth.

Real-time Context Engine for Business Moment



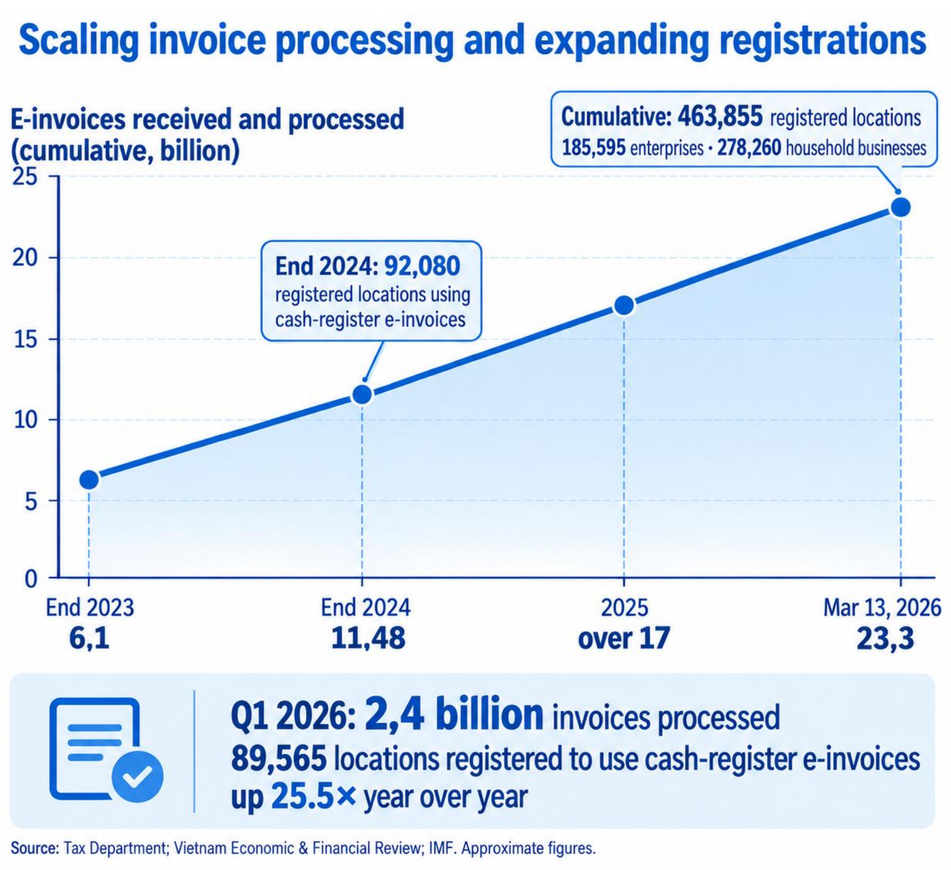
Case Study: eInvoice VN

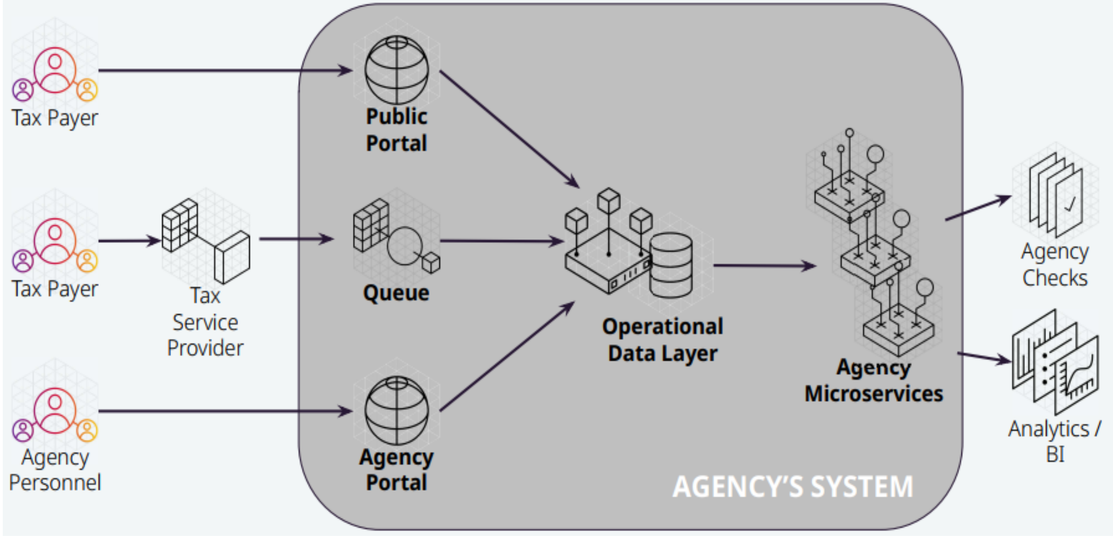
Challenge
Customer needed a scalable invoice aggregation platform for collecting invoice information from tax payers as well as from around 30 government related service providers. Information is used for analysis and finding fraudulent cases.

Solution
For the mission critical and high load on-premises needs, they have selected distributed database (IBM DataStax) as the scalable, highly-available data platform, including additional functionalities for analysis and investigation.

Value
Ability to collect and process billions of invoices per year, the phase 1 target was 15 billion invoices in 2022

https://tapchikinhtetaichinh.vn/3-thang-dau-nam-2026-tiep-nhan-va-xu-ly-24-ty-hoa-don-dien-tu-152991.html

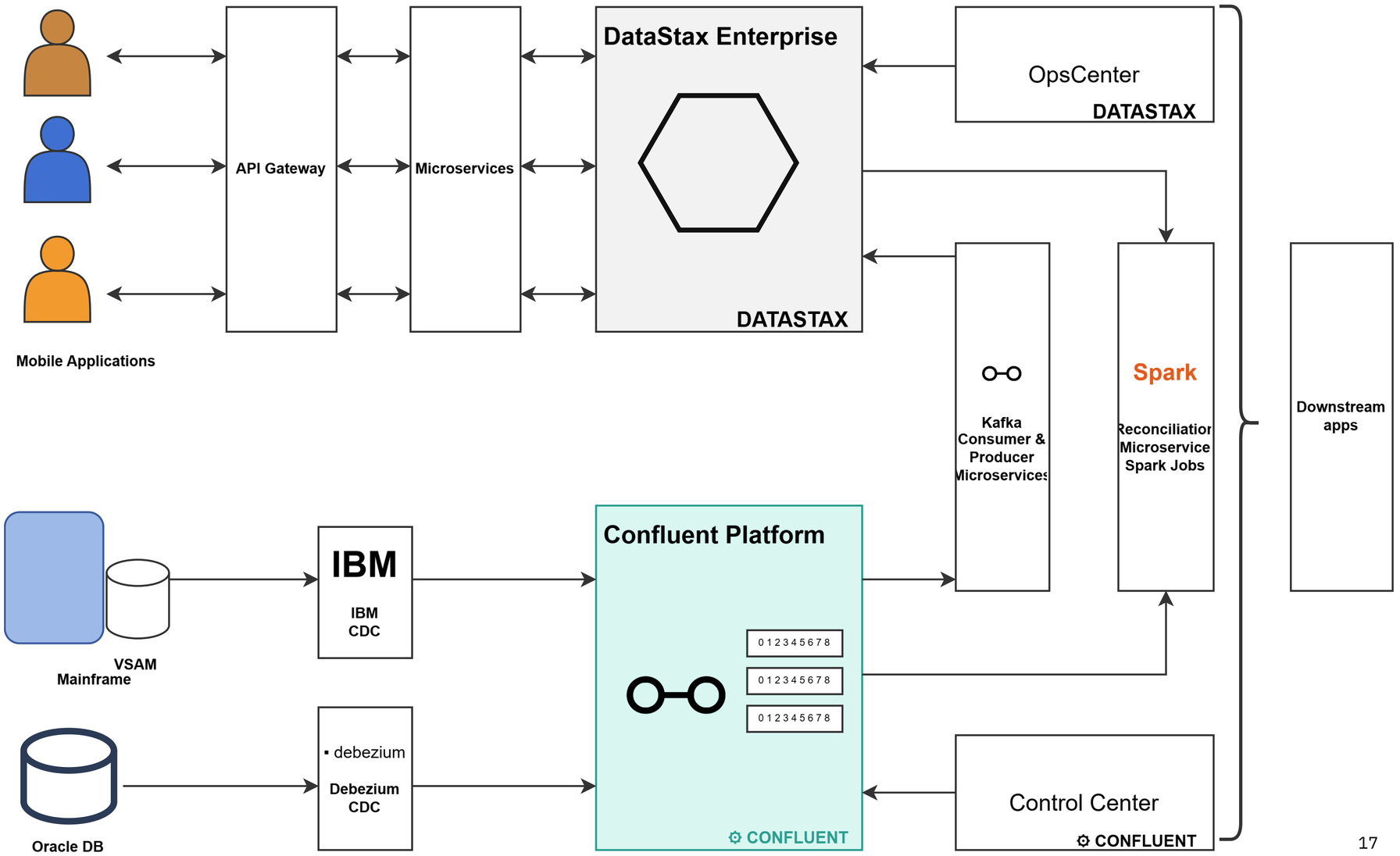




Case Study: Large Bank in Thailand

Context

Customer embarked on modernizing their platform to cater to needs of better and new services while reducing processing latency and impact on the system of records which was reaching its operational capacity to serve future services.



Thank you!

“Transactions happen in milliseconds, and AI decisions need to happen just as fast.” **Rob Thomas, IBM**







